

Virtualization Points

1 Virtualization Fundamentals

1. Virtualization is the process of creating virtual versions of physical resources.
2. Virtualization allows multiple operating systems to run on one physical machine.
3. Virtualization improves hardware utilization.
4. Virtualization reduces hardware cost.
5. Virtualization simplifies server management.
6. Virtualization improves scalability and flexibility.
7. Virtual machines are isolated from each other.
8. Virtualization supports testing and development environments.
9. Virtualization enables faster deployment of servers.
10. Virtualization reduces power and cooling costs.

2 Hypervisor

1. Hypervisor is software that creates and manages virtual machines.
2. Hypervisor controls CPU, memory, and storage allocation.
3. Type 1 hypervisor runs directly on hardware.
4. Type 2 hypervisor runs on top of an operating system.
5. Type 1 hypervisor is also called bare-metal hypervisor.
6. Type 2 hypervisor is also called hosted hypervisor.
7. Examples of Type 1: VMware ESXi, Microsoft Hyper-V, Xen.
8. Examples of Type 2: VMware Workstation, VirtualBox.
9. Hypervisor provides isolation between virtual machines.

3 Virtual Machine Components

1. Virtual Machine (VM) is a software-based computer.
2. VM has virtual CPU, virtual RAM, and virtual disk.
3. VM uses virtual network adapters.
4. VM runs its own operating system.
5. VM files are stored as image files.
6. VM can be started, stopped, and restarted.
7. VM snapshot captures VM state.
8. VM cloning creates a copy of a virtual machine.

4 Virtualization Storage Concepts

1. Datastore is storage used to store VM files.
2. VM disk is stored as VMDK file.
3. Storage can be local or shared.
4. Shared storage supports VM migration.
5. SAN provides block-level storage.
6. NAS provides file-level storage.
7. iSCSI is used for SAN over IP network.
8. Fibre Channel is high-speed storage network.
9. Storage virtualization improves storage utilization.

5 Virtual Networking

1. Virtual Switch connects virtual machines.
2. VM network adapter connects VM to virtual switch.
3. Port group defines network configuration.
4. VLAN is used to separate network traffic.
5. Virtual networking allows VM communication.
6. Bridged networking connects VM to physical network.
7. NAT networking allows VM to access internet through host.
8. Host-only networking allows VM communication with host only.

6 VM Migration and High Availability

1. VM migration moves VM from one host to another.
2. Live migration moves VM without downtime.
3. vMotion is VMware live migration technology.
4. Storage vMotion migrates VM storage.
5. High Availability (HA) restarts VM if host fails.
6. Fault Tolerance (FT) provides continuous availability.
7. Cluster is a group of hosts working together.
8. HA improves system reliability.

7 Resource Management

1. CPU and memory are shared between VMs.
2. Resource pool is used to manage resource allocation.
3. CPU overcommit allows more VMs than physical CPU.
4. Memory overcommit allows more VMs than physical RAM.
5. Reservation guarantees minimum resources.

6. Limit restricts maximum resource usage.
7. Shares define priority during resource contention.

8 Virtualization Benefits

1. Server consolidation reduces hardware usage.
2. Isolation improves security.
3. Snapshots help in backup and testing.
4. Easy VM backup and restore.
5. Disaster recovery becomes easier.
6. Testing can be done without affecting production.
7. Multiple OS can run on one system.

9 Virtualization Challenges

1. Requires high initial setup cost.
2. Requires shared storage for advanced features.
3. Performance overhead may occur.
4. Requires skilled administrators.
5. VM sprawl can create management issues.
6. Backup and storage management is important.